

Influence of environmental and population factors on Prince William Sound spawning phenology.

Bia Dias

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Models

PWS model results

```
PWSdata <- read.csv("DATA_v2/fulldata.csv")

glm_list_full <- list(
glm_full <- glm(DOY_cum50~ SSB+Mean_age+ ICOADS.SST.apr.wmean + Upwell.wi+
               NCEP.MerWind.fa+ a.PNA.su, data=PWSdata),

glm_full_a <- glm(DOY_cum50~ Mean_age+ ICOADS.SST.apr.wmean + Upwell.wi+
                 NCEP.MerWind.fa+ a.PNA.su, data=PWSdata),

glm_full_b<- glm(DOY_cum50~ Mean_age+ ICOADS.SST.apr.wmean + Upwell.wi+
                 a.PNA.su, data=PWSdata),
glm_full_c<- glm(DOY_cum50~ ICOADS.SST.apr.wmean + Upwell.wi+ a.PNA.su,
                 data=PWSdata),
glm_full_d<- glm(DOY_cum50~ SSB, data=PWSdata),
glm_full_e<- glm(DOY_cum50~ Mean_age, data=PWSdata),
glm_full_f<- glm(DOY_cum50~ ICOADS.SST.apr.wmean, data=PWSdata),
glm_full_g<- glm(DOY_cum50~ Upwell.wi, data=PWSdata),
glm_full_h<- glm(DOY_cum50~ a.PNA.su, data=PWSdata),
glm_full_i <- glm(DOY_cum50~ Mean_age+ ICOADS.SST.mar.wmean + Upwell.wi+
                 NCEP.MerWind.fa+ a.PNA.su, data=PWSdata))
```

Table 1: Table continues below

	Models	LL	K	AICc
2	~mean age + April SST + winter downwelling+ Fall meridional winds+ Summer PNA	-121.965	7	261.429
3	~mean age + April SST + winter downwelling+Summer PNA	-124.741	6	264.026

	Models	LL	K	AICc
1	~SSB + mean age + April SST + winter downwelling+ Fall meridional winds+ Summer PNA	-121.694	8	264.032
4	~April SST + winter downwelling + Summer PNA	-126.47	5	264.704
10	~mean age + March SST + winter downwelling+ Fall meridional winds+ Summer PNA	-123.649	7	264.798
9	Summer PNA	-132.696	3	272.058
8	winter downwelling	-133.964	3	274.594
5	~SSB	-133.966	3	274.598
6	~mean age	-134.415	3	275.498
7	~April SST	-134.683	3	276.034

	$\Delta AICc$	AICc wt.	ER	D^2
2	0	0.517	1	0.521
3	2.597	0.1411	3.664	0.449
1	2.603	0.1407	3.674	0.527
4	3.275	0.1006	5.141	0.4
10	3.369	0.0959	5.389	0.479
9	10.629	0.0025	203.266	0.18
8	13.165	7e-04	722.265	0.127
5	13.169	7e-04	723.625	0.127
6	14.068	5e-04	1134.664	0.107
7	14.604	3e-04	1483.29	0.095

```

## Response variable: DOY_cum50
## Total response variance: 55.76346
## Analysis based on 40 observations
##
## 5 Regressors:
## Mean_age ICOADS.SST.apr.wmean Upwell.wi NCEP.MerWind.fa a.PNA.su
## Proportion of variance explained by model: 52.07%
## Metrics are normalized to sum to 100% (rela=TRUE).
##
## Relative importance metrics:
##
##                               lmg
## Mean_age                    0.1591259
## ICOADS.SST.apr.wmean      0.1468749
## Upwell.wi                  0.1747282
## NCEP.MerWind.fa           0.1821159
## a.PNA.su                   0.3371551
##
## Average coefficients for different model sizes:
##
##                               1X          2Xs          3Xs          4Xs
## Mean_age                    3.17825468  2.98493037  2.79845469  2.61784050
## ICOADS.SST.apr.wmean      -3.22615730 -3.04359020 -2.88040811 -2.73827530
## Upwell.wi                  -0.01961574 -0.01820732 -0.01676177 -0.01530569

```

```
## NCEP.MerWind.fa      1.67104122  1.59345276  1.52065384  1.44885273
## a.PNA.su             -5.59768394 -5.57233324 -5.54301555 -5.50612975
##                      5Xs
## Mean_age            2.44203323
## ICOADS.SST.apr.wmean -2.61860352
## Upwell.wi           -0.01386428
## NCEP.MerWind.fa      1.37468218
## a.PNA.su             -5.45925406
```

Western model results

```
data_W <- read.csv("DATA_v2/fulldata_W.csv",header = T)

glm_list_W <- list(
  glm_W_1<- glm(DOY_cum50~ SSB, data=data_W),
  glm_W_2<- glm(DOY_cum50~ MDM_area, data=data_W),
  glm_W_3<- glm(DOY_cum50~ NCEP.SST.fa,
                data=data_W),
  glm_W_4<- glm(DOY_cum50~ NCEP.SST.wi, data=data_W),
  glm_W_5<- glm(DOY_cum50~ NCEP.SST.sp, data=data_W),
  glm_W_6<- glm(DOY_cum50~ Upwell.wi, data=data_W),
  glm_W_7<- glm(DOY_cum50~ a.NPGO.wi, data=data_W),
  glm_W_8<- glm(DOY_cum50~ a.NPGO.su, data=data_W),

  glm_W_9<- glm(DOY_cum50~ SSB + ICOADS.SST.apr.int+
                NCEP.SST.sp+Upwell.wi+a.NPGO.wi, data=data_W),

  glm_W_10<- glm(DOY_cum50~ SSB + ICOADS.SST.apr.int+
                 Upwell.wi+a.NPGO.su, data=data_W),

  glm_W_11<- glm(DOY_cum50~ SSB + ICOADS.SST.apr.int+
                 Upwell.wi+a.NPGO.wi, data=data_W),

  glm_W_12 <- glm(DOY_cum50~ NCEP.SST.sp+ NCEP.SST.wi+ ICOADS.SST.apr.int,
                 data=data_W),

  glm_W_13 <- glm(DOY_cum50~ FWinput.total.t+ ICOADS.SST.apr.int,
                 data=data_W),

  glm_W_14 <- glm(DOY_cum50~ MDM_area+FWinput.total.t+ ICOADS.SST.apr.int,
                 data=data_W),

  glm_W_15 <- glm(DOY_cum50~ SSB+Mean_age+ NCEP.MerWind.wi+
                 FWinput.total.t+ ICOADS.SST.apr.int,
                 data=data_W),

  glm_W_16 <- glm(DOY_cum50~ Mean_age+
                 FWinput.total.t+ ICOADS.SST.apr.int,
                 data=data_W),

  glm_W_17 <- glm(DOY_cum50~ MDM_area+Mean_age+
                 FWinput.total.t+ ICOADS.SST.apr.int,
```

```

        data=data_W),

glm_W_18 <- glm(DOY_cum50~ Mean_age+ NCEP.MerWind.wi+
        FWinput.total.t+ ICOADS.SST.apr.int,
        data=data_W),

glm_W_19 <- glm(DOY_cum50~ Mean_age+ NCEP.MerWind.wi+
        FWinput.total.t+ ICOADS.SST.apr.int+
        Upwell.wi+NCEP.MerWind.fa+ a.PNA.su,
        data=data_W),

glm_W_20 <- glm(DOY_cum50~ Mean_age+
        FWinput.total.t+ ICOADS.SST.apr.int+
        Upwell.wi+NCEP.MerWind.fa+ a.PNA.su,
        data=data_W),

glm_W_21 <- glm(DOY_cum50~ Mean_age+
        FWinput.total.t+a.PNA.su,
        data=data_W), ## best model

glm_W_22 <- glm(DOY_cum50~ Mean_age+
        FWinput.total.t+a.MEI.v2.su,
        data=data_W),

glm_W_23 <- glm(DOY_cum50~ Mean_age, data=data_W),

glm_W_24 <- glm(DOY_cum50~
        FWinput.total.t+a.PNA.su,
        data=data_W)

)

```

Table 3: Table continues below

	Models	LL	K	AICc
21	~mean age + modeled freshwater input + Summer PNA	-92.629	5	197.757
16	~mean age + modeled freshwater input+ April SST	-94.705	5	201.909
22	~mean age + modeled freshwater input + Summer MEI.v2	-95.108	5	202.716
23	~mean age	-98.234	3	203.391
18	~mean age + Winter meridional winds + modeled freshwater input + April SST	-94.639	6	204.929
17	~Mile-days of Milt + mean age + modeled freshwater input + April SST	-94.703	6	205.058
20	~mean age + modeled freshwater input + April SST + winter downwelling + Fall meridional winds + Summer PNA	-91.385	8	205.627
15	~SSB+ mean age + Winter meridional winds + modeled freshwater input + April SST	-94.621	7	208.333

	Models	LL	K	AICc
19	~mean age + Winter meridional winds + modeled freshwater input + April SST + winter downwelling + Fall meridional winds + Summer PNA	-90.901	9	208.801
13	~modeled freshwater input+ April SST	-120.798	4	250.887
24	~modeled freshwater input+Summer PNA	-120.939	4	251.169
14	~Mile-days of Milt + modeled freshwater input+ April SST	-120.554	5	253.108
5	~Spring offshore SST	-123.352	3	253.454
2	~Mile-days of Milt	-123.519	3	253.788
1	~SSB	-123.56	3	253.869
4	~Winter offshore SST	-123.635	3	254.021
8	~Summer NPGO	-123.702	3	254.153
3	~Fall offshore SST	-123.745	3	254.241
6	~winter downwelling	-123.778	3	254.306
7	~Winter NPGO	-123.779	3	254.308
12	~Spring offshore SST+ Winter offshore SST + April SST	-122.946	5	257.892
11	~SSB + April SST + winter downwelling + Winter NPGO	-122.943	6	260.783
10	~SSB + April SST + winter downwelling+ Summer NPGO	-122.945	6	260.786
9	~SSB + April SST + Spring offshore SST + winter downwelling + Winter NPGO	-122.703	7	263.405

	$\Delta AICc$	AICc wt.	ER	D^2
21	0	0.74	1	0.375
16	4.152	0.0928	7.973	0.283
22	4.959	0.062	11.935	0.263
23	5.634	0.0442	16.724	0.092
18	7.172	0.0205	36.095	0.286
17	7.301	0.0192	38.496	0.283
20	7.87	0.0145	51.158	0.425
15	10.576	0.0037	197.917	0.287
19	11.044	0.003	250.134	0.443
13	53.13	0	344324243578.433	0.153
24	53.412	0	396460787520.899	0.146
14	55.351	0	1045459963349.17	0.164
5	55.697	0	1242944911259.74	0.024
2	56.031	0	1468820031198	0.015
1	56.112	0	1529769413942.74	0.012
4	56.264	0	1650062213629.07	0.008
8	56.396	0	1763245128960.11	0.004
3	56.484	0	1842062847748.48	0.002
6	56.549	0	1903471424605.48	0
7	56.551	0	1904641199794.77	0
12	60.134	0	11429684450150.8	0.045
11	63.026	0	48511630382071.8	0.046
10	63.029	0	48589055516105.8	0.045
9	65.648	0	180012771762363	0.058

```

## Response variable: DOY_cum50
## Total response variance: 46.61609
## Analysis based on 30 observations
##
## 3 Regressors:
## Mean_age FWinput.total.t a.PNA.su
## Proportion of variance explained by model: 37.54%
## Metrics are normalized to sum to 100% (rela=TRUE).
##
## Relative importance metrics:
##
##                               lmg
## Mean_age                0.3872591
## FWinput.total.t 0.4299410
## a.PNA.su                0.1827999
##
## Average coefficients for different model sizes:
##
##                1X          2Xs          3Xs
## Mean_age      -2.456933 -3.024470 -3.892420
## FWinput.total.t -5.141861 -6.401905 -8.219609
## a.PNA.su      -2.182880 -2.852429 -4.065171

```

Eastern model results

```

data_E <- read.csv("DATA_v2/fulldata_E.csv", header = T)

glm_list_E <- list(
  glm_E_1<- glm(DOY_cum50~ SSB, data=data_E),
  glm_E_2<- glm(DOY_cum50~ MDM_area, data=data_E),
  glm_E_3<- glm(DOY_cum50~ ICOADS.SST.apr.int, data=data_E),
  glm_E_4<- glm(DOY_cum50~ NCEP.SST.fa,
    data=data_E),
  glm_E_5<- glm(DOY_cum50~ NCEP.SST.wi, data=data_E),
  glm_E_6<- glm(DOY_cum50~ NCEP.SST.sp, data=data_E),
  glm_E_7<- glm(DOY_cum50~ Upwell.wi, data=data_E),
  glm_E_8<- glm(DOY_cum50~ a.PNA.su, data=data_E),
  glm_E_9<- glm(DOY_cum50~ a.NPGO.su, data=data_E),
  glm_E_10<- glm(DOY_cum50~ NCEP.MerWind.wi, data=data_E),
  glm_E_11<- glm(DOY_cum50~ NCEP.MerWind.fa, data=data_E),
  glm_E_12<- glm(DOY_cum50~ NCEP.ZonWind.wi, data=data_E),
  glm_E_13<- glm(DOY_cum50~ NCEP.ZonWind.sp, data=data_E),

  glm_E_14<- glm(DOY_cum50~ SSB +
    NCEP.ZonWind.wi, data=data_E),

  glm_E_15<- glm(DOY_cum50~ SSB + ICOADS.SST.apr.int, data=data_E),

  glm_E_16<- glm(DOY_cum50~ SSB + NCEP.MerWind.wi, data=data_E),

  glm_E_17<- glm(DOY_cum50~ SSB + NCEP.MerWind.fa, data=data_E),

  glm_E_18<- glm(DOY_cum50~ SSB + NCEP.ZonWind.sp, data=data_E),

```

```

glm_E_19<- glm(DOY_cum50~ SSB + ICOADS.SST.apr.int+
               NCEP.ZonWind.wi, data=data_E),
glm_E_20 <- glm(DOY_cum50~ FWinput.total.t,data=data_E),

glm_E_21 <- glm(DOY_cum50~ SSB + NCEP.MerWind.wi +NCEP.ZonWind.wi+Upwell.wi+
               a.PNA.su, data=data_E),

glm_E_22 <- glm(DOY_cum50~ MDM_area + NCEP.MerWind.wi +Upwell.wi+
               a.PNA.su, data=data_E),

glm_E_23 <- glm(DOY_cum50~ SSB + NCEP.MerWind.wi +Upwell.wi, data=data_E),

glm_E_24 <- glm(DOY_cum50~ Mean_age+NCEP.MerWind.wi +Upwell.wi,
               data=data_E),

glm_E_25 <- glm(DOY_cum50~ Mean_age+
               FWinput.total.t+ ICOADS.SST.apr.int+
               Upwell.wi+NCEP.MerWind.fa+ a.PNA.su,
               data=data_E),

glm_E_26 <- glm(DOY_cum50~ MDM_area + Mean_age+
               NCEP.MerWind.wi+ Upwell.wi, data=data_E),

glm_E_27 <- glm(DOY_cum50~ MDM_area + Mean_age+
               ICOADS.SST.apr.int+
               NCEP.MerWind.wi+ Upwell.wi+a.PNA.su, data=data_E),

glm_E_28 <- glm(DOY_cum50~ MDM_area + NCEP.MerWind.wi +Upwell.wi+
               a.PNA.wi, data=data_E),

glm_E_29 <- glm(DOY_cum50~ MDM_area + NCEP.MerWind.fa +Upwell.wi+
               a.PNA.su, data=data_E)

)

```

Table 5: Table continues below

	Models	LL	K	AICc
22	~mile-days of milt + Winter meridional winds + Winter downwelling + Summer PNA	-124.546	6	263.638
26	~mile-days of milt + mean age + Winter meridional winds + Winter downwelling	-126.236	6	267.017
28	~mile-days of milt + Winter meridional winds + Winter downwelling + Winter PNA	-126.293	6	267.132
23	~SSB + Winter meridional winds + Winter downwelling	-128.482	5	268.728
24	~mean age + Winter meridional winds + Winter downwelling	-128.519	5	268.803
29	~mile-days of milt + Fall meridional winds + Winter downwelling + Summer PNA	-127.139	6	268.824
10	~ Winter meridional wind	-131.39	3	269.447

	Models	LL	K	AICc
27	~mile-days of milt + mean age+ICOADS.SST.apr.int+ Winter meridional winds + Winter downwelling + Summer PNA	-124.43	8	269.504
21	~SSB + Winter meridional winds + Winter zonal wind + Winter downwelling + Summer PNA	-126.185	7	269.87
14	~SSB + Winter zonal wind	-130.603	4	270.348
16	~SSB + Winter meridional wind	-130.788	4	270.72
1	~SSB	-132.909	3	272.485
19	~SSB + April SST + Winter zonal wind	-130.45	5	272.665
17	~SSB + Fall meridional wind	-132.032	4	273.207
2	~Mile-days of Milt	-133.352	3	273.371
18	~SSB + Spring zonal wind	-132.249	4	273.642
15	~SSB + April SST	-132.289	4	273.721
11	~Fall meridional wind	-134.195	3	275.056
7	~Winter downwelling	-135.706	3	278.079
3	~April SST	-135.961	3	278.59
25	~mean age + FWinput.total.t + April SST + Winter downwelling + Fall meridional winds + Summer PNA	-129.069	8	278.784
8	~Summer PNA	-136.514	3	279.694
13	~Spring zonal wind	-137.363	3	281.392
12	~Winter zonal wind	-137.553	3	281.773
4	~Fall offshore SST	-137.834	3	282.334
5	~Winter offshore SST	-138.124	3	282.915
6	~Spring offshore SST	-138.184	3	283.034
9	~Summer NPGO	-138.191	3	283.048
20	~modeled freshwater input	-138.197	3	283.061

	$\Delta AICc$	AICc wt.	ER	D^2
22	0	0.537	1	0.495
26	3.379	0.0991	5.417	0.45
28	3.494	0.0936	5.738	0.449
23	5.09	0.0421	12.745	0.385
24	5.166	0.0406	13.234	0.384
29	5.186	0.0402	13.372	0.425
10	5.81	0.0294	18.263	0.288
27	5.867	0.0286	18.792	0.498
21	6.232	0.0238	22.559	0.452
14	6.711	0.0187	28.656	0.316
16	7.082	0.0156	34.5	0.31
1	8.848	0.0064	83.42	0.232
19	9.027	0.0059	91.253	0.321
17	9.569	0.0045	119.643	0.265
2	9.733	0.0041	129.871	0.215
18	10.004	0.0036	148.716	0.257
15	10.083	0.0035	154.731	0.256
11	11.419	0.0018	301.699	0.181
7	14.442	4e-04	1367.726	0.117

	$\Delta AICc$	AICc wt.	ER	D^2
3	14.952	3e-04	1765.069	0.106
25	15.146	3e-04	1945.157	0.366
8	16.056	2e-04	3065.951	0.081
13	17.755	1e-04	7167.166	0.041
12	18.136	1e-04	8672.481	0.032
4	18.697	0	11479.379	0.018
5	19.277	0	15345.414	0.004
6	19.397	0	16290.926	0.001
9	19.41	0	16400.678	0
20	19.424	0	16510.604	0

```
## Response variable: DOY_cum50
## Total response variance: 60.17885
## Analysis based on 40 observations
##
## 4 Regressors:
## MDM_area NCEP.MerWind.wi Upwell.wi a.PNA.su
## Proportion of variance explained by model: 49.47%
## Metrics are normalized to sum to 100% (rela=TRUE).
##
## Relative importance metrics:
##
##                               lmg
## MDM_area           0.2657920
## NCEP.MerWind.wi    0.3824777
## Upwell.wi          0.2292840
## a.PNA.su           0.1224463
##
## Average coefficients for different model sizes:
##
##                1X          2Xs          3Xs          4Xs
## MDM_area      0.12427482  0.10510638  0.08777291  0.07255127
## NCEP.MerWind.wi 2.57354743  2.29112942  2.00326740  1.71647470
## Upwell.wi     -0.01959081 -0.01940403 -0.01921971 -0.01907323
## a.PNA.su      -3.89110276 -3.41497844 -3.13066760 -3.01837354
```

PWS Western Eastern model results

```
dataPWS <- read_csv(file="DATA_v2/fullldata_PWS.csv")

## Rows: 80 Columns: 40

## -- Column specification -----
## Delimiter: ","
## chr  (1): Region
## dbl (39): Year, DOY_Start, DOY_End, DOY_cum25, DOY_cum50, DOY_cum75, N, Cond...

##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.

dataPWS$RS_SSB_10 <- as_factor(dataPWS$RS_SSB_10)
dataPWS$RS_SSB_5  <- as_factor(dataPWS$RS_SSB_5)
dataPWS$RS_SP_10  <- as_factor(dataPWS$RS_SP_10)
```

```

dataPWS$RS_SP_5 <- as_factor(dataPWS$RS_SP_5)
dataPWS$Region <- as_factor(dataPWS$Region)

dataPWS_v2 <- dataPWS %>% drop_na()
dataPWS_v2$Region <- factor(dataPWS_v2$Region, levels = rev(levels(dataPWS_v2$Region)))

#Second interaction of model selection

glm_list1 <- list(
  glm_1_1 <- glm(DOY_cum50~ MDM_area+ NCEP.MerWind.wi +Upwell.wi+
    a.PNA.su, data=dataPWS_v2),

  glm_1_2 <- glm(DOY_cum50~ Mean_age+
    FWinput.total.t+a.PNA.su,
    data=dataPWS_v2),

  glm_1_3 <- glm(DOY_cum50~ MDM_area*Region +Mean_age*Region+
    NCEP.MerWind.wi +Upwell.wi+
    FWinput.total.t*Region+
    a.PNA.su,
    data=dataPWS_v2),

  glm_1_4 <- glm(DOY_cum50~ Region+MDM_area*Region +Mean_age*Region+
    NCEP.MerWind.wi +Upwell.wi+
    FWinput.total.t+
    a.PNA.su,
    data=dataPWS_v2), ##

  glm_1_5 <- glm(DOY_cum50~ Region+MDM_area*Region*RS_SSB_10 +Mean_age*Region+
    NCEP.MerWind.wi +Upwell.wi+
    FWinput.total.t+
    a.PNA.su,
    data=dataPWS_v2),

  glm_1_6 <- glm(DOY_cum50~ Region+ RS_SSB_10+MDM_area*RS_SSB_10+
    Mean_age*RS_SSB_10+
    NCEP.MerWind.wi +Upwell.wi+
    FWinput.total.t+
    a.PNA.su,
    data=dataPWS_v2),

  glm_1_7 <- glm(DOY_cum50~ RS_SSB_10+
    Mean_age*RS_SSB_10+
    NCEP.MerWind.wi +Upwell.wi+
    FWinput.total.t+
    a.PNA.su,
    data=dataPWS_v2),

  glm_1_8 <- glm(DOY_cum50~
    FWinput.total.t+a.PNA.su,
    data=dataPWS_v2),

```

```

glm_1_9 <- glm(DOY_cum50~
  FWinput.total.t*RS_SSB_10,
  data=dataPWS_v2),

glm_1_10 <- glm(DOY_cum50~
  Mean_age*RS_SSB_10,
  data=dataPWS_v2),

glm_1_11 <- glm(DOY_cum50~ Region+MDM_area*Region + Mean_age*RS_SSB_10+
  NCEP.MerWind.wi +Upwell.wi+
  FWinput.total.t+
  a.PNA.su,
  data=dataPWS_v2),

glm_1_12 <- glm(DOY_cum50~ Region+MDM_area*Region +
  NCEP.MerWind.wi +Upwell.wi+
  FWinput.total.t*Region+
  a.PNA.su,
  data=dataPWS_v2),

glm_WE_13 <- glm(DOY_cum50~ Region+MDM_area*Region +Mean_age*Region+
  NCEP.MerWind.wi + ICOADS.SST.apr.int+ICOADS.SST.mar.intp +Upwell.wi+
  FWinput.total.t+
  a.PNA.su,
  data=dataPWS),

glm_WE_14 <- glm(DOY_cum50~ Region+MDM_area*Region +Mean_age*Region+
  NCEP.MerWind.wi + ICOADS.SST.apr.int +Upwell.wi+
  FWinput.total.t+
  a.PNA.su,
  data=dataPWS),

glm_WE_15 <- glm(DOY_cum50~ Region+MDM_area*Region +Mean_age*Region+
  NCEP.MerWind.wi + ICOADS.SST.mar.intp +Upwell.wi+
  FWinput.total.t+
  a.PNA.su,
  data=dataPWS),

glm_1_0 <- glm(DOY_cum50~1, data=dataPWS_v2),
glm_1_17 <- glm(DOY_cum50~ Region+MDM_area*Region*RS_SSB_5 +Mean_age*Region+
  NCEP.MerWind.wi +Upwell.wi+
  FWinput.total.t+
  a.PNA.su,
  data=dataPWS_v2),

glm_1_18 <- glm(DOY_cum50~ Region+MDM_area*Region*RS_SP_10 +Mean_age*Region+
  NCEP.MerWind.wi +Upwell.wi+
  FWinput.total.t+
  a.PNA.su,
  data=dataPWS_v2),

glm_1_19 <- glm(DOY_cum50~ Region+MDM_area*Region*RS_SP_5 +Mean_age*Region+
  NCEP.MerWind.wi +Upwell.wi+

```

```

FWinput.total.t+
a.PNA.su,
data=dataPWS_v2)
)

```

```

## Warning in aictab.AICglm.lm(glm_list1, modnames = c("~mile-days of milt + Winter meridional wind + W
## Check model structure carefully as some models may be redundant

## [1] 0.2263049
## [1] 488.7612
## [1] 0.2001999
## [1] 488.6882
## [1] 0.5704866
## [1] 463.7048
## [1] 0.5702209
## [1] 460.8261
## [1] 0.602485
## [1] 467.7006
## [1] 0.4461664
## [1] 481.4996
## [1] 0.364588
## [1] 482.6446
## [1] 0.1956665
## [1] 486.7618
## [1] 0.3483534
## [1] 474.3481
## [1] 0.1556177
## [1] 492.4853
## [1] 0.523195
## [1] 471.0166
## [1] 0.4984962
## [1] 468.807
## [1] 0.5755077
## [1] 465.908
## [1] 0.571022
## [1] 463.6175
## [1] 0.5752952
## [1] 462.9167
## [1] 0
## [1] 497.5674
## [1] 0.6384529
## [1] 475.3729
## [1] 0.6058343
## [1] 467.1083
## [1] 0.6384529
## [1] 475.3729

```

Table 7: Table continues below

	Models	LL	K	AICc
4	~region + mile-days of milt <i>region + mean age</i> Region + Winter meridional winds + Winter downwelling + modeled freshwater input + Summer PNA	-217.137	11	460.826

	Models	LL	K	AICc
15	~region + mile-days of milt <i>region + mean</i> <i>ageregion</i> + Winter meridional winds+ March SST + Winter downwelling + modeled freshwater input + Summer PNA	-216.722	12	462.917
14	~region + mile-days of milt <i>region + mean</i> <i>ageregion</i> + Winter meridional winds+ April SST + Winter downwelling + modeled freshwater input + Summer PNA	-217.072	12	463.617
3	~mile-days of milt <i>Region + mean ageregion</i> + Winter meridional winds + Winter downwelling + modeled freshwater input* <i>Region</i> + Summer PNA	-217.116	12	463.705
13	~region + mile-days of milt <i>region + mean</i> <i>ageregion</i> + Winter meridional winds+ April SST + March SST + Winter downwelling + modeled freshwater input + Summer PNA	-216.704	13	465.908
18	~region+ Surplus production 10-year regime shift + mile-days of milt * Surplus production 10-year regime shift + mean age* Surplus production 10-year regime shift + Winter meridional winds + Winter downwelling + modeled freshwater input + Summer PNA	-214.11	15	467.108
5	~region + mile-days of milt <i>region</i> SSB 10-year regime shift +mean age* <i>region</i> + Winter meridional winds + Winter downwelling + modeled freshwater input + Summer PNA	-214.406	15	467.701
12	~region + mile-days of milt * <i>region</i> + Winter meridional winds + Winter downwelling + modeled freshwater input + Summer PNA	-222.539	10	468.807
11	~region + mile-days of milt <i>region + mean</i> <i>age</i> SSB 10-year regime shift + Winter meridional winds + Winter downwelling + modeled freshwater input + Summer PNA	-220.771	12	471.017
9	~modeled freshwater input* SSB 10-year regime shift	-231.705	5	474.348
17	~region+ SSB 5-year regime shift + mile-days of milt <i>SSB 5-year regime shift +</i> <i>mean age</i> SSB 5-year regime shift + Winter meridional winds + Winter downwelling + modeled freshwater input + Summer PNA	-211.086	19	475.373
19	~region+ Surplus production 5-year regime shift + mile-days of milt * Surplus production 5-year regime shift + mean age* Surplus production 5-year regime shift + Winter meridional winds + Winter downwelling + modeled freshwater input + Summer PNA	-211.086	19	475.373

	Models	LL	K	AICc
6	~region+ SSB 10-year regime shift + mile-days of milt <i>SSB 10-year regime shift</i> + <i>mean age</i> SSB 10-year regime shift + Winter meridional winds + Winter downwelling + modeled freshwater input + Summer PNA	-226.013	12	481.5
7	~SSB 10-year regime shift + mean age* SSB 10-year regime shift + Winter meridional winds + Winter downwelling + modeled freshwater input + Summer PNA	-230.822	9	482.645
8	~modeled freshwater input	-239.073	4	486.762
2	~mean age + modeled freshwater input + Summer PNA	-238.875	5	488.688
1	~mile-days of milt + Winter meridional wind + Winter downwelling + Summer PNA	-237.714	6	488.761
10	~mean age*SSB 10-year regime shift	-240.774	5	492.485
16	~null model	-246.694	2	497.567

	$\Delta AICc$	AICc wt.	ER	D^2
4	0	0.4956	1	0.57
15	2.091	0.1742	2.844	0.575
14	2.791	0.1227	4.038	0.571
3	2.879	0.1175	4.218	0.57
13	5.082	0.0391	12.691	0.576
18	6.282	0.0214	23.129	0.606
5	6.874	0.0159	31.101	0.602
12	7.981	0.0092	54.079	0.498
11	10.191	0.003	163.245	0.523
9	13.522	6e-04	863.486	0.348
17	14.547	3e-04	1441.404	0.638
19	14.547	3e-04	1441.404	0.638
6	20.673	0	30845.144	0.446
7	21.818	0	54678.134	0.365
8	25.936	0	428405.835	0.196
2	27.862	0	1122485.981	0.2
1	27.935	0	1164190.285	0.226
10	31.659	0	7493746.8	0.156
16	36.741	0	95118135.237	0

```
## Response variable: DOY_cum50
## Total response variance: 68.36874
## Analysis based on 70 observations
##
## 9 Regressors:
## Region MDM_area Mean_age NCEP.MerWind.wi Upwell.wi FWinput.total.t a.PNA.su Region:MDM_area Region:M
## Proportion of variance explained by model: 57.02%
## Metrics are normalized to sum to 100% (rela=TRUE).
##
## Relative importance metrics:
```

```
##
##                               lmg
## Region                0.27901340
## MDM_area              0.03789226
## Mean_age              0.01560841
## NCEP.MerWind.wi      0.07366974
## Upwell.wi            0.13260379
## FWinput.total.t      0.11974039
## a.PNA.su             0.09883463
## Region:MDM_area      0.14197767
## Region:Mean_age      0.10065971
##
## Average coefficients for different model sizes:
##
##               1X           2Xs           3Xs           4Xs           5Xs
## Region        -7.70833333 -8.66735961 -10.60222749 -14.25530852 -19.482762552
## MDM_area       0.04680292  0.03800408  0.02759323  0.01635068  0.005966811
## Mean_age      -0.45140864 -0.66036209 -0.91947278 -1.36069568 -1.980797087
## NCEP.MerWind.wi 1.36528444  1.26866617  1.19278789  1.13901566  1.107334848
## Upwell.wi     -0.01966260 -0.01894392 -0.01825486 -0.01757230 -0.017014108
## FWinput.total.t 3.13727471  1.92897325  0.77307980 -0.77123572 -2.705609719
## a.PNA.su      -3.60084745 -3.37921512 -3.27550926 -3.24302347 -3.294564948
## Region:MDM_area      NaN          NaN    0.14220543  0.14510048  0.145909645
## Region:Mean_age      NaN          NaN    2.94083894  3.67026524  4.171650340
##
##               6Xs           7Xs           8Xs           9Xs
## Region        -25.735629253 -32.74220678 -40.53586671 -49.03319608
## MDM_area       -0.002542052  -0.01015120  -0.01825205  -0.02731194
## Mean_age      -2.606536793  -3.14251843  -3.69704914  -4.42824184
## NCEP.MerWind.wi 1.086491435  1.06641624  1.04258058  1.00601657
## Upwell.wi     -0.016736218  -0.01669026  -0.01667558  -0.01654616
## FWinput.total.t -4.359772803  -5.28989287  -5.77983422  -6.15803059
## a.PNA.su      -3.406094512  -3.53063867  -3.68465257  -3.89415714
## Region:MDM_area  0.144059146  0.13995847  0.13503962  0.13074200
## Region:Mean_age  4.431237293  4.55550615  4.70163073  4.96199841
```

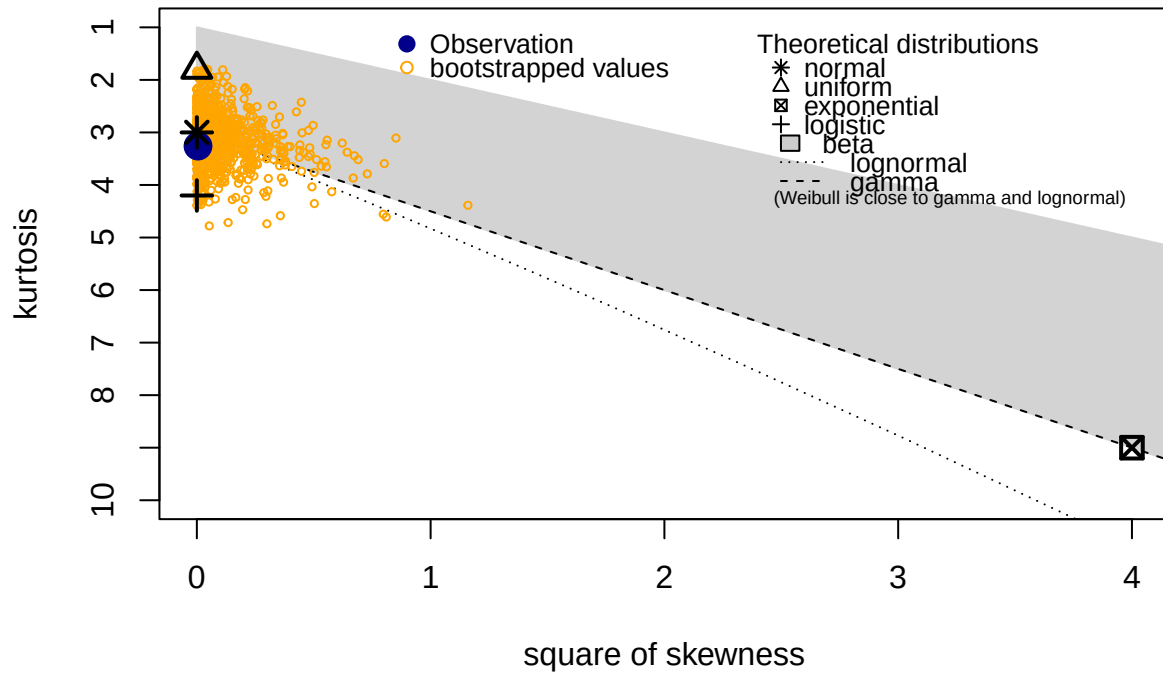
Model diagnostics

- Collinearity: measured by the variance inflation factor (VIF), which is:

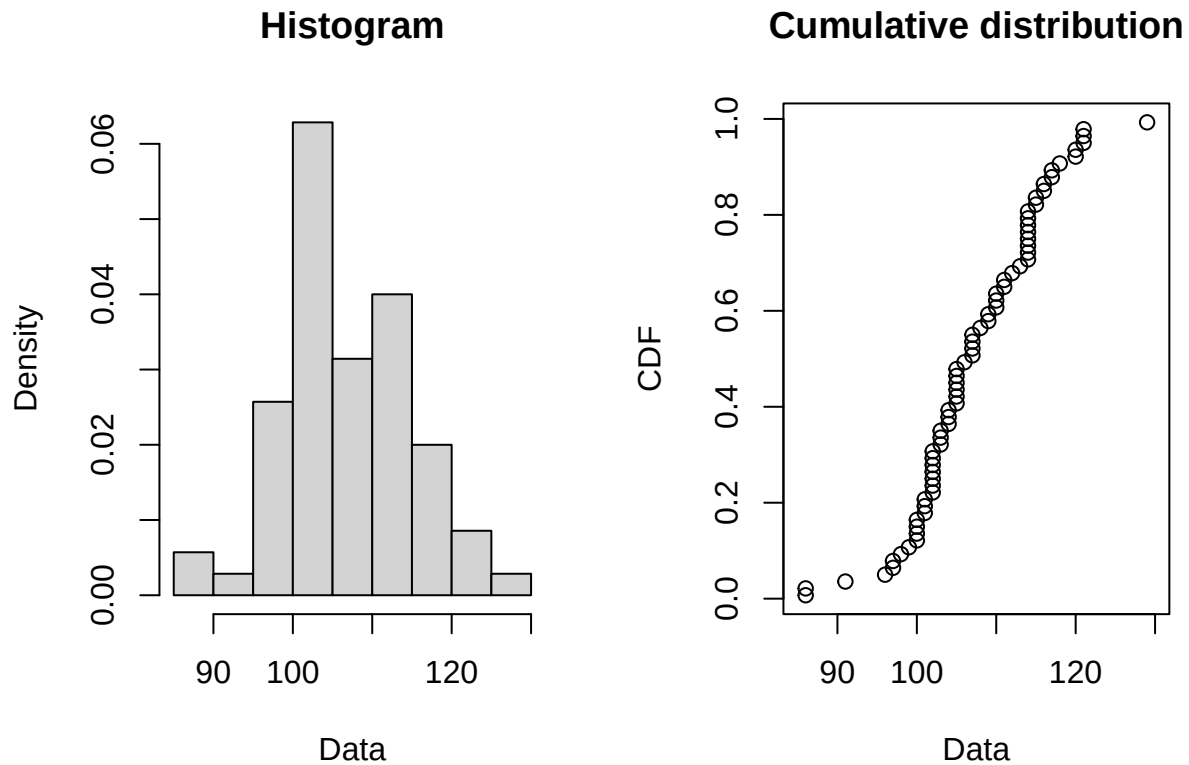
$$VIF = \frac{1}{1 - R^2}$$

```
# Distribution (Packages MASS and fitdistrplus) Cullen and
# Frey graph
descdist(dataPWS_v2$DOY_cum50, boot = 1000)
```

Cullen and Frey graph



```
## summary statistics
## -----
## min: 86   max: 129
## median: 106.5
## mean: 107.3286
## estimated sd: 8.268539
## estimated skewness: -0.07024019
## estimated kurtosis: 3.259834
plotdist(dataPWS_v2$DOY_cum50)
```

*# discrete since we are dealing with day. the distribution
is moderately skewed.*

```
lm_full_a_PWS <- lm(DOY_cum50 ~ Mean_age + ICOADS.SST.apr.wmean +
  Upwell.wi + NCEP.MerWind.fa + a.PNA.su, data = PWSdata)
ols_coll_diag(lm_full_a_PWS)
```

Tolerance and Variance Inflation Factor

```
## -----
##           Variables Tolerance      VIF
## 1           Mean_age 0.9607796 1.040821
## 2 ICOADS.SST.apr.wmean 0.9808035 1.019572
## 3           Upwell.wi 0.8896708 1.124011
## 4      NCEP.MerWind.fa 0.9003429 1.110688
## 5             a.PNA.su 0.9881872 1.011954
##
##
```

Eigenvalue and Condition Index

```
## -----
##      Eigenvalue Condition Index  intercept  Mean_age ICOADS.SST.apr.wmean
## 1 3.960566929      1.000000 3.999128e-04 1.266386e-03 6.949374e-04
## 2 1.001808021      1.988321 1.581725e-06 1.818581e-05 2.292462e-06
## 3 0.885103687      2.115347 1.301810e-04 4.737144e-04 2.352647e-04
## 4 0.129528816      5.529622 3.277564e-03 9.122452e-03 7.771902e-03
## 5 0.018770025     14.526007 8.282957e-03 6.763788e-01 2.167168e-01
## 6 0.004222523     30.626168 9.879078e-01 3.127405e-01 7.745788e-01
##      Upwell.wi NCEP.MerWind.fa      a.PNA.su
```

```
## 1 9.343068e-03    0.0085881010 1.470040e-07
## 2 8.977077e-05    0.0072361062 9.758731e-01
## 3 4.335043e-05    0.8713381432 1.108287e-02
## 4 9.695828e-01    0.1063330335 4.408146e-03
## 5 9.746804e-03    0.0060301695 7.662314e-03
## 6 1.119418e-02    0.0004744465 9.734693e-04
```

```
ols_correlations(lm_full_a_PWS)
```

```
##                      Correlations
## -----
## Variable              Zero Order   Partial   Part
## -----
## Mean_age              0.327        0.335     0.246
## ICOADS.SST.apr.wmean  -0.308        -0.337    -0.247
## Upwell.wi             -0.356        -0.324    -0.237
## NCEP.MerWind.fa       0.342         0.360     0.267
## a.PNA.su              -0.425        -0.511    -0.412
## -----
```

```
ols_test_breusch_pagan(lm_full_a_PWS, rhs = T, multiple = T) #heteroscedasticity test
```

```
##
## Breusch Pagan Test for Heteroskedasticity
## -----
## Ho: the variance is constant
## Ha: the variance is not constant
##
##                      Data
## -----
## Response : DOY_cum50
## Variables: Mean_age ICOADS.SST.apr.wmean Upwell.wi NCEP.MerWind.fa a.PNA.su
##
##          Test Summary (Unadjusted p values)
## -----
## Variable              chi2        df        p
## -----
## Mean_age              0.37370862     1    0.54098966
## ICOADS.SST.apr.wmean  0.01796494     1    0.89337620
## Upwell.wi             1.05540215     1    0.30426620
## NCEP.MerWind.fa       4.19960579     1    0.04043338
## a.PNA.su              3.64210708     1    0.05633559
## -----
## simultaneous          10.88387619     5    0.05373161
## -----
```

```
lm_W_21 <- lm(DOY_cum50 ~ Mean_age + FWinput.total.t + a.PNA.su,
  data = data_W)
```

```
ols_coll_diag(lm_W_21)
```

```
## Tolerance and Variance Inflation Factor
## -----
##          Variables Tolerance      VIF
## 1          Mean_age 0.8995294 1.111692
```

```
## 2 FWinput.total.t 0.8810976 1.134948
## 3      a.PNA.su 0.9281670 1.077392
##
##
## Eigenvalue and Condition Index
## -----
##      Eigenvalue Condition Index      intercept      Mean_age FWinput.total.t
## 1 3.00944508      1.000000 6.240588e-04 0.0025222851      1.117096e-03
## 2 0.96272683      1.768038 5.048955e-05 0.0001229211      5.382689e-05
## 3 0.02417309      11.157764 6.661807e-03 0.5823718264      1.562193e-01
## 4 0.00365500      28.694558 9.926636e-01 0.4149829673      8.426097e-01
##      a.PNA.su
## 1 0.0055374710
## 2 0.9260428988
## 3 0.0003839508
## 4 0.0680356794

ols_correlations(lm_W_21)

##                      Correlations
## -----
## Variable              Zero Order      Partial      Part
## -----
## Mean_age              -0.304         -0.500         -0.457
## FWinput.total.t       -0.320         -0.519         -0.480
## a.PNA.su              -0.187         -0.391         -0.336
## -----

ols_test_breusch_pagan(lm_W_21, rhs = T, multiple = T) #heteroscedasticity test

##
## Breusch Pagan Test for Heteroskedasticity
## -----
## Ho: the variance is constant
## Ha: the variance is not constant
##
##                      Data
## -----
## Response : DOY_cum50
## Variables: Mean_age FWinput.total.t a.PNA.su
##
##      Test Summary (Unadjusted p values)
## -----
##      Variable              chi2      df      p
## -----
##      Mean_age              2.038388      1      0.15337209
##      FWinput.total.t       2.686215      1      0.10121998
##      a.PNA.su              2.732858      1      0.09830321
## -----
##      simultaneous          7.394486      3      0.06033244
## -----

lm_E_22 <- lm(DOY_cum50 ~ MDM_area + NCEP.MerWind.wi + Upwell.wi +
  a.PNA.su, data = data_E)

ols_coll_diag(lm_E_22)
```

```
## Tolerance and Variance Inflation Factor
```

```
## -----
```

```
##      Variables Tolerance      VIF
## 1      MDM_area 0.7557989 1.323103
## 2 NCEP.MerWind.wi 0.7644604 1.308112
## 3      Upwell.wi 0.9814499 1.018901
## 4      a.PNA.su 0.9689090 1.032089
```

```
##
```

```
##
```

```
## Eigenvalue and Condition Index
```

```
## -----
```

```
##      Eigenvalue Condition Index      intercept      MDM_area NCEP.MerWind.wi
## 1      2.9394012          1.000000 0.013199166 0.027168068      0.03224474
## 2      1.0305647          1.688853 0.002338868 0.001350205      0.02727506
## 3      0.6990247          2.050610 0.019270054 0.001458977      0.64197748
## 4      0.2511399          3.421145 0.002892644 0.727229204      0.24152795
## 5      0.0798694          6.066514 0.962299269 0.242793545      0.05697476
##      Upwell.wi      a.PNA.su
## 1 0.016818694 0.0006354835
## 2 0.004777548 0.8511726928
## 3 0.027005337 0.1176865089
## 4 0.178290893 0.0299522970
## 5 0.773107529 0.0005530178
```

```
ols_correlations(lm_E_22)
```

```
##      Correlations
```

```
## -----
```

```
## Variable      Zero Order      Partial      Part
## -----
## MDM_area      0.464      0.314      0.235
## NCEP.MerWind.wi 0.537      0.403      0.313
## Upwell.wi      -0.342      -0.421      -0.330
## a.PNA.su      -0.284      -0.292      -0.217
```

```
## -----
```

```
ols_test_breusch_pagan(lm_E_22, rhs = T, multiple = T) #heteroscedasticity test
```

```
##
```

```
## Breusch Pagan Test for Heteroskedasticity
```

```
## -----
```

```
## Ho: the variance is constant
```

```
## Ha: the variance is not constant
```

```
##
```

```
##      Data
```

```
## -----
```

```
## Response : DOY_cum50
```

```
## Variables: MDM_area NCEP.MerWind.wi Upwell.wi a.PNA.su
```

```
##
```

```
##      Test Summary (Unadjusted p values)
```

```
## -----
```

```
##      Variable      chi2      df      p
## -----
## MDM_area      1.29866467      1      0.2544573
## NCEP.MerWind.wi 1.80669706      1      0.1789050
```

##	Upwell.wi	0.02931299	1	0.8640584
##	a.PNA.su	0.08401979	1	0.7719221
##	-----			
##	simultaneous	2.19497450	4	0.6999494
##	-----			